



Customer Shopping Behavior Analysis

Analysis of 3,900 transactions to reveal spending patterns, segments, product preferences, and subscription behavior to guide strategy.

Dataset at a Glance

Scope

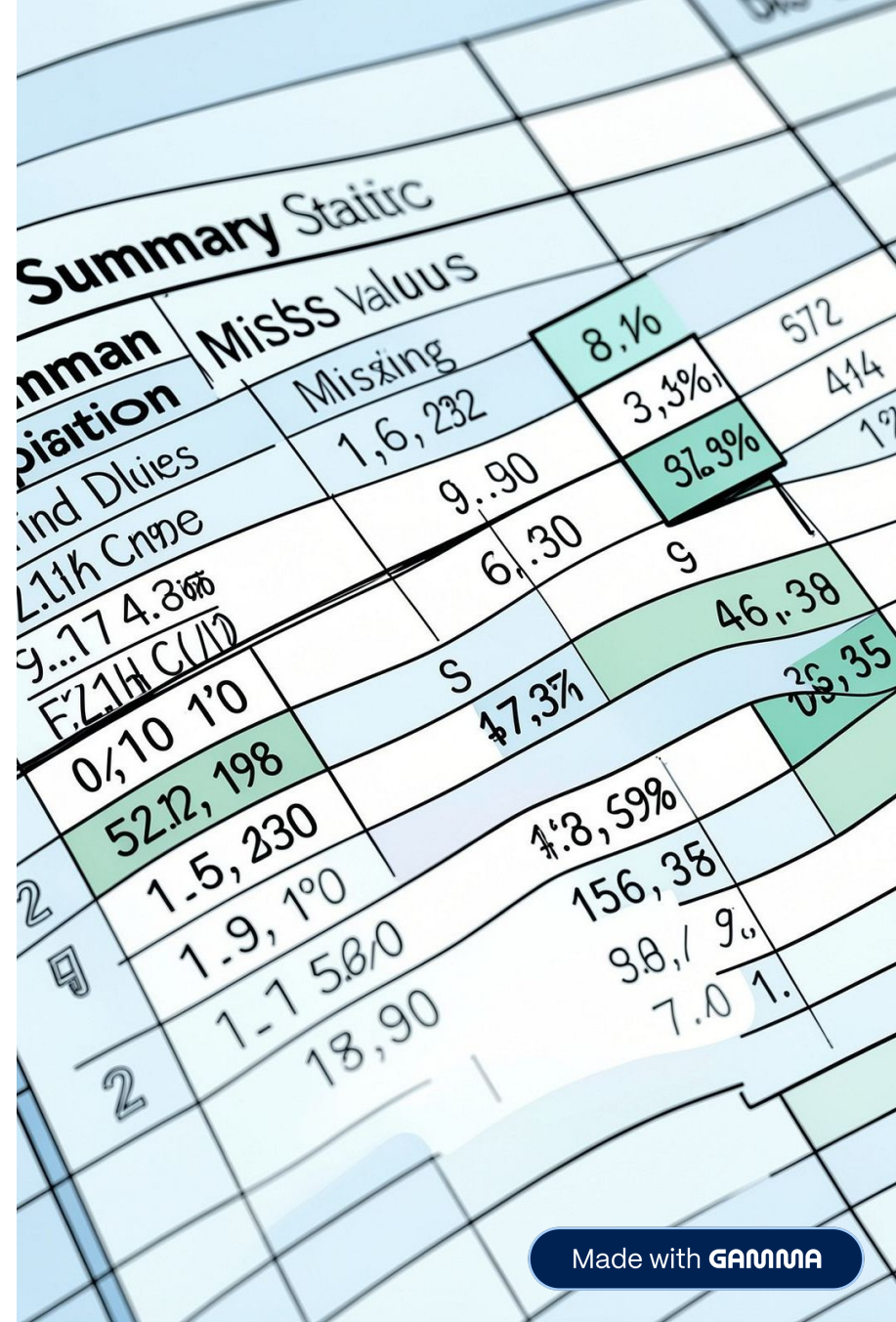
3,900 rows × 18 columns — customer demographics, purchase details, and shopping behavior.

Key Features

Age, Gender, Location, Item, Category, Purchase Amount, Season, Rating, Subscription, Shipping, Discounts.

Data Quality

37 missing values in Review Rating column — handled during cleaning.



Data Preparation & Feature Engineering



Loaded with pandas; used `df.info()` and `.describe()`.

- Imputed Review Rating by category median.
- Renamed columns to snake_case.
- Engineered age_group and purchase_frequency_days.
- Dropped promo_code_used after redundancy check with discount_applied.
- Exported cleaned table to PostgreSQL for SQL analysis.

High-Level Statistics

3.9K

Customers

Total transactions / unique customers.

\$59.76

Avg Purchase

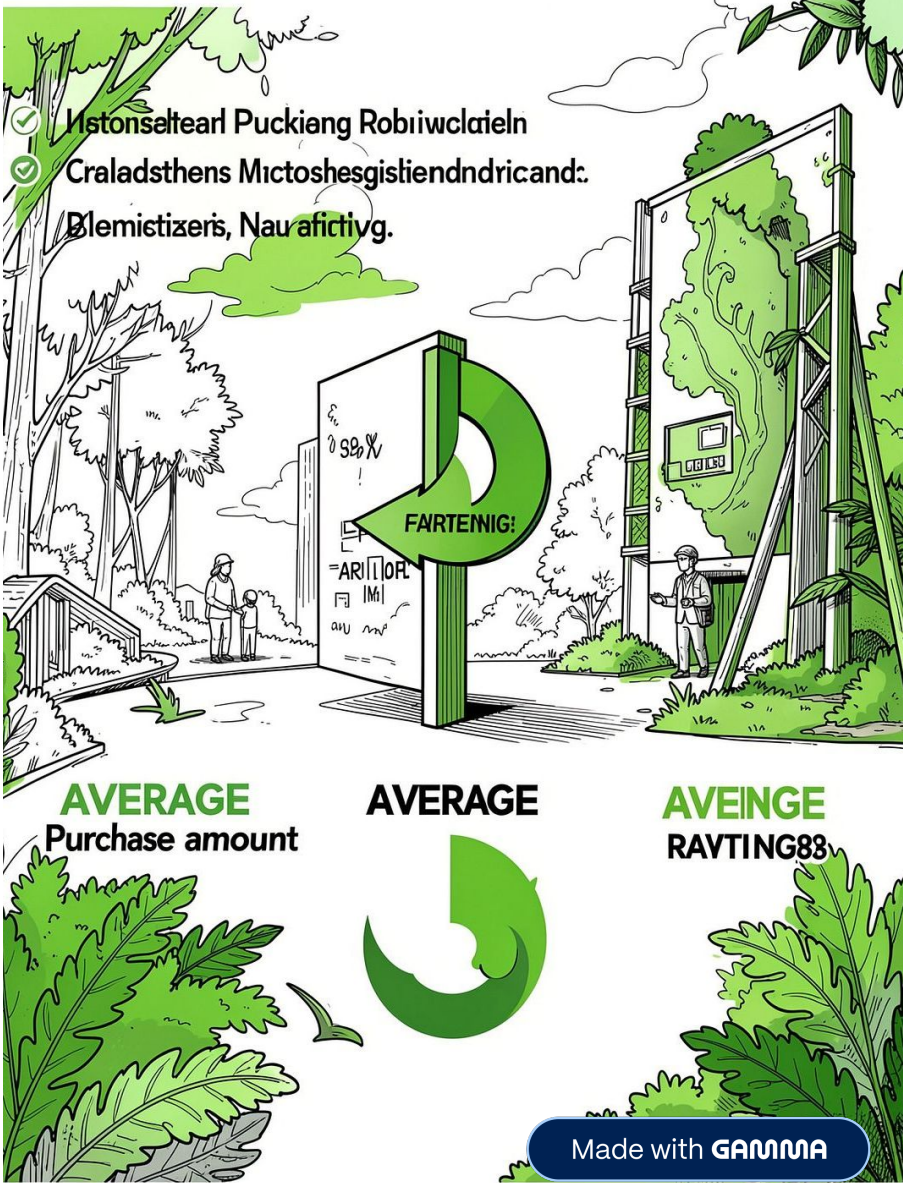
Average purchase amount across dataset.

3.75

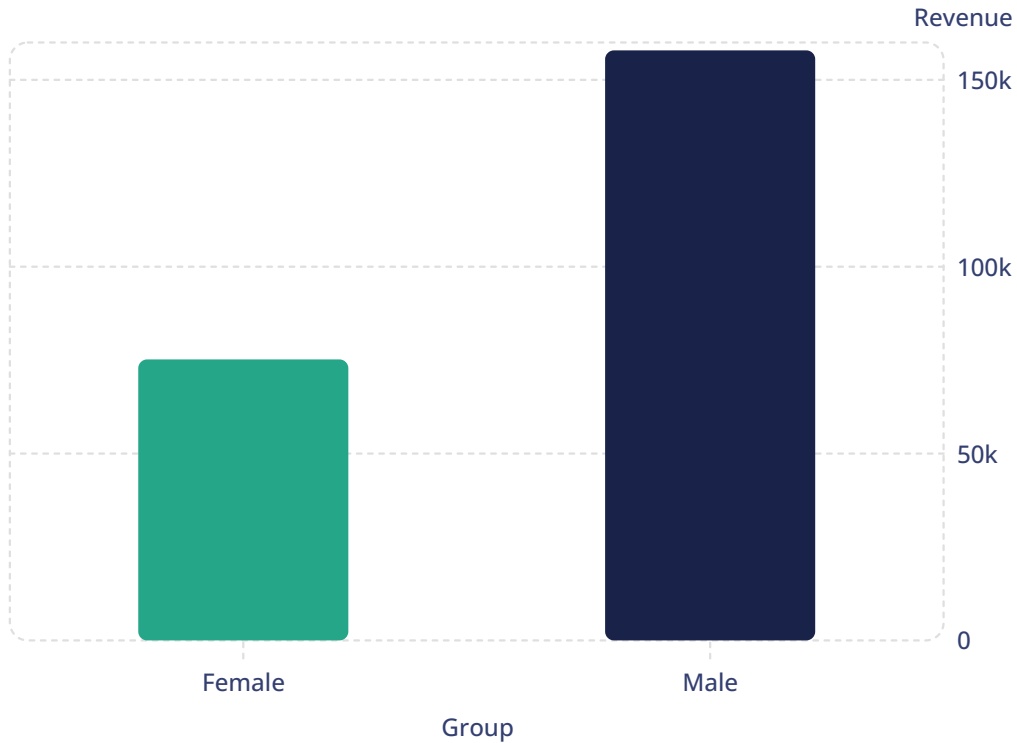
Avg Rating

Average review rating (after imputation).

Customer Feedback Metrics



Revenue & Spending Insights (SQL)



Male customers generated higher total revenue. SQL queries also identified high-spending discount users (839 rows) who used discounts yet spent above average.

Top Products & Discount Dependence



Top Rated

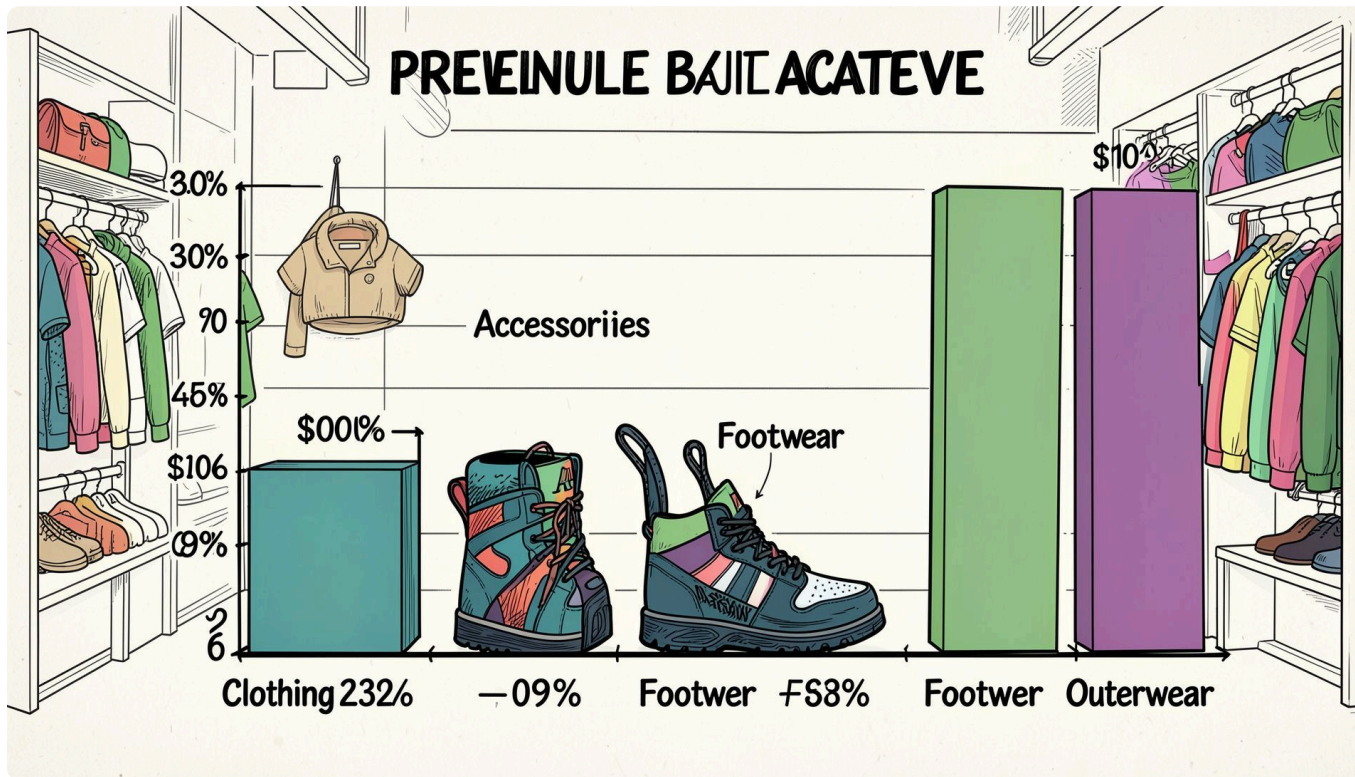
Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78)



Discount-Dependent

Highest discount rates: Hat 50.00%, Sneakers 49.66%, Coat 49.07%, Sweater 48.17%, Pants 47.37%.

Category Performance & Top Items

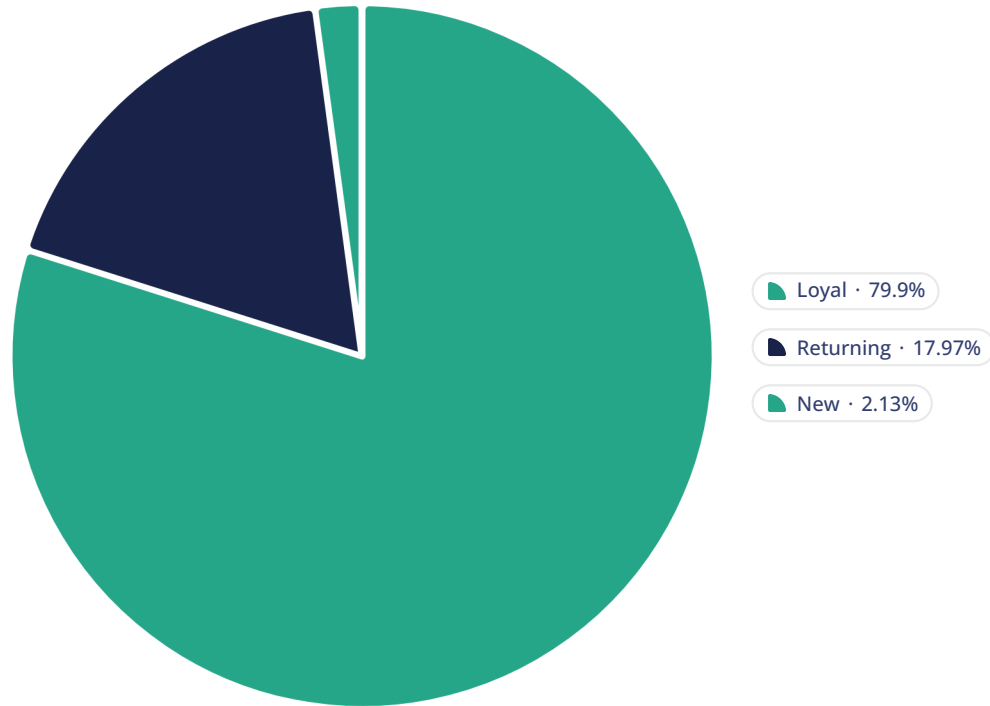


Top items by category (most orders):

- Accessories: Jewelry, Sunglasses, Belt
- Clothing: Blouse, Pants, Shirt
- Footwear: Sandals, Shoes, Sneakers
- Outerwear: Jacket, Coat

Clothing drives the largest revenue and sales volume.

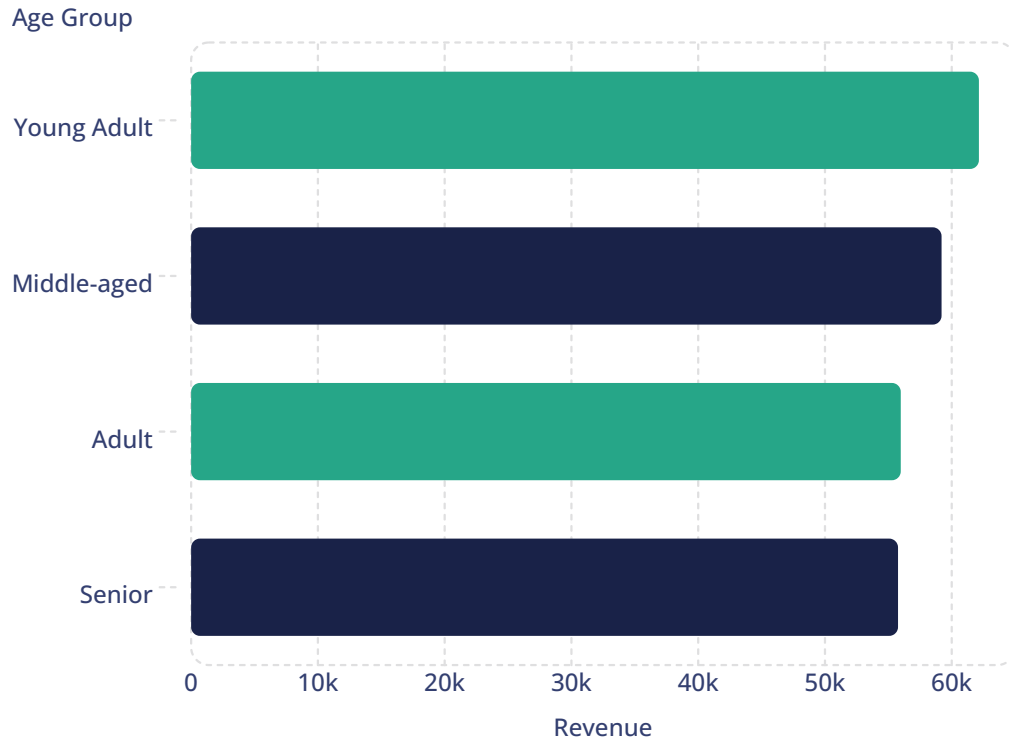
Customer Segments & Behavior



Segments defined by purchase history: Loyal customers dominate (3,116). Repeat buyers: 958 subscribers vs. 2,518 non-subscribers among repeat buyers.

Subscribers (1,053) vs Non-subscribers (2,847): avg spend ~ \$59.5 and total revenue shown in SQL analysis.

Age & Shipping Insights

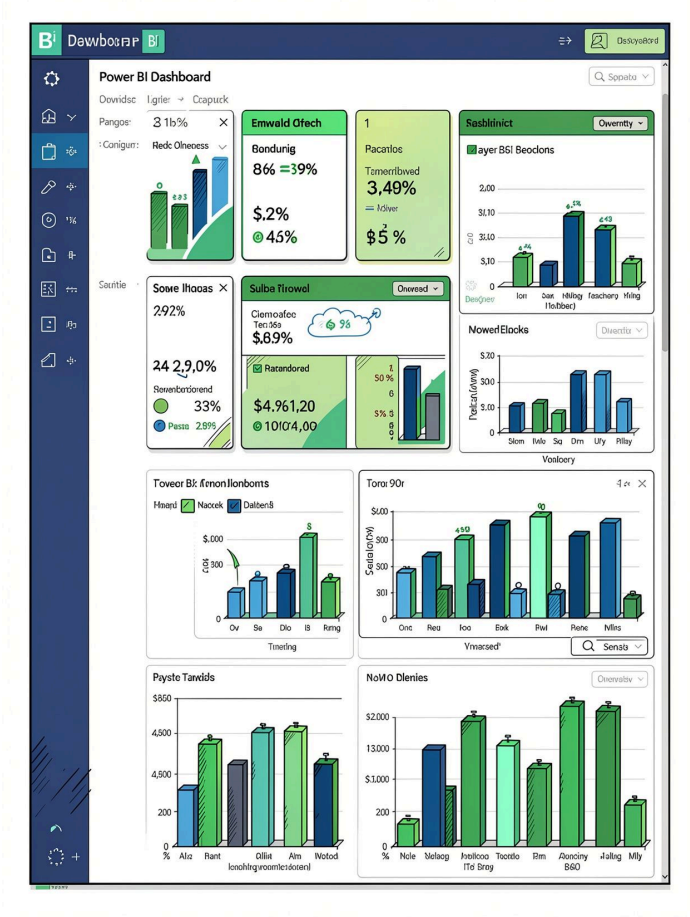


Revenue by age group: Young Adults lead. Shipping comparison: Express average \$60.48 vs Standard \$58.46 — both similar but Express slightly higher.

Recommendations & Dashboard



Interactive Power BI dashboard summarizes metrics: 3.9K customers, \$59.76 avg purchase, 3.75 avg rating, revenue and sales by category and age group — supports these recommendations.



Boost Subscriptions

Promote exclusive subscriber benefits to increase recurring revenue.

Loyalty Programs

Reward repeat buyers to grow the Loyal segment.

Review Discounts

Balance discount-driven sales with margin controls for dependent products.

Product Positioning

Highlight top-rated and best-selling items in campaigns.

Targeted Marketing

Focus on high-revenue age groups and express-shipping customers.